SCHEME and SOLUTIONS

Date: 13-01-26

Subject: Theory of Computations

Sub. Code: CS354TA

Improvement CIE

PART-A

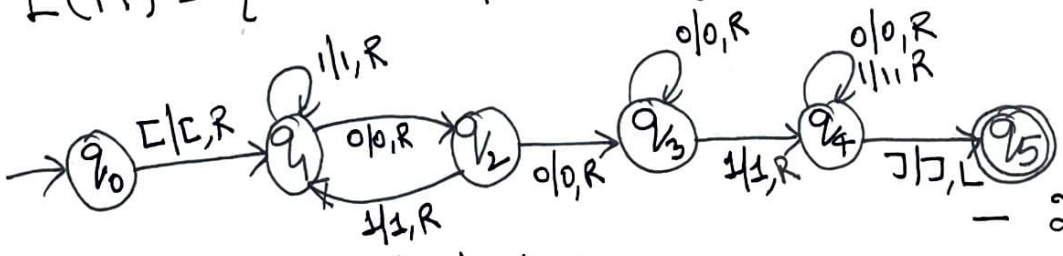
1.1

— 2 marks

— 2 marks

1.2 $L(M) = \{ a^n b^n a^m \mid n, m \geq 1 \}$

1.3



— 2 marks

(Or) Any equivalent LBA.

1.4 $S \to XY$

$X \to ACX \mid \epsilon$

$Y \to bY \mid \epsilon$

$CA \to AC$

$Cb \to bC$

$bC \to bc$

$cC \to cc$

$A \to a$

(Or) Any other equivalent grammar

Even we can write CFG for this language. If students writes CFG marks to be alloted. — 1 Mark

Ex: $S \to aSc \mid B \mid \epsilon$

$B \to bB \mid \epsilon$

$S \Rightarrow XY \Rightarrow ACXY \Rightarrow ACACXY \Rightarrow ACACY \Rightarrow AACCY \Rightarrow AACCbY$

$\Rightarrow AACCbbY \Rightarrow AAC\underline{Cb}b \Rightarrow AA\underline{Cb}Cb \Rightarrow AAbC\underline{Cb}$

$\Rightarrow AAb\underline{Cb}C \Rightarrow AAbbCC \Rightarrow aAbbCC \Rightarrow aabbCC$ — 1 Mark

$\Rightarrow aabbcC \Rightarrow aabbcc$

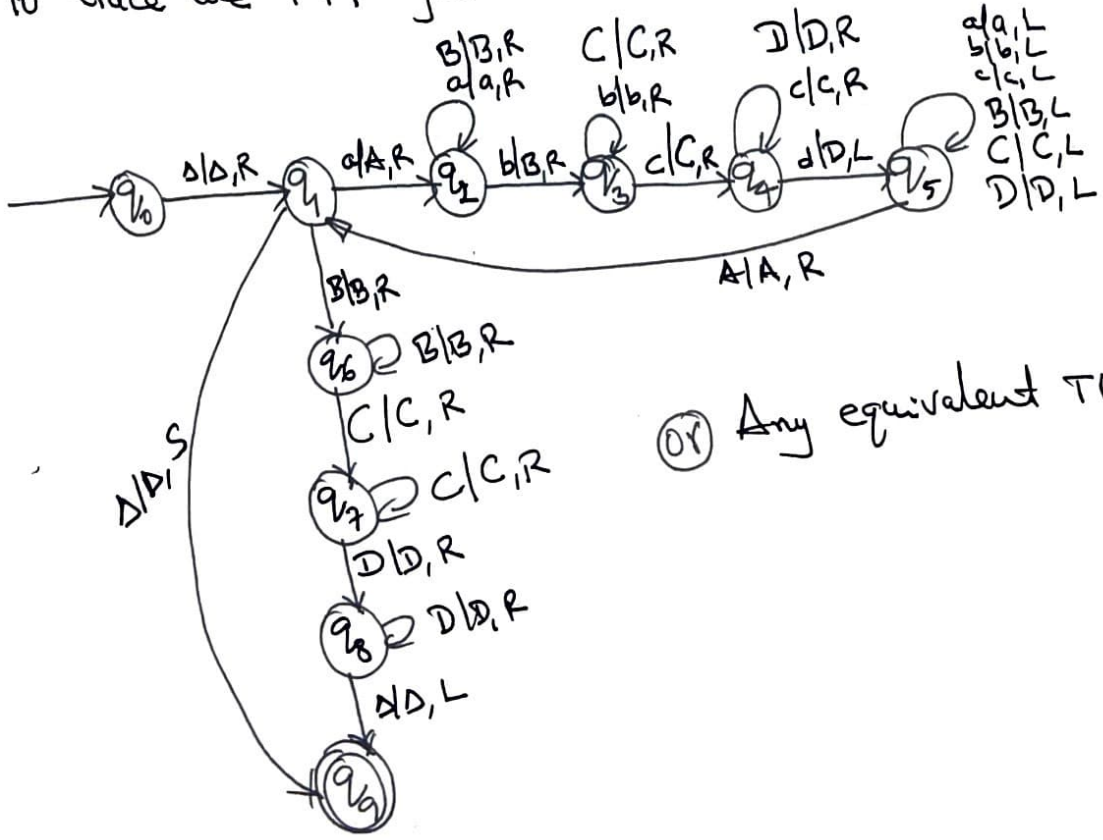
1.5 $(q_0, \underline{\Delta}aabcba\Delta) \vdash (q_1, \Delta\underline{a}abcba\Delta) \vdash (q_2, \Delta a\underline{a}bcba\Delta)$
$\vdash (q_2, \Delta aa\underline{b}cba\Delta) \vdash (q_3, \Delta aab\underline{c}ba\Delta) \vdash (q_4, \Delta aabc\underline{b}a\Delta)$
$\vdash (q_5, \Delta aabcb\underline{a}\Delta) \vdash (q_6, \Delta aabcba\underline{\Delta}) \vdash (q_7, \Delta aabcb\underline{a}\Delta)$

The last ID is $(q_7, \Delta aabcb\underline{a}\Delta)$

If the students write only the last ID give 1 mark.
If the complete process is shown – 2 marks.

PART – B

1. TM definition — 1 mark
Language of TM definition — 1 mark
TM Construction for the given language — 4 marks
To trace the TM for the aabbccdd — 4 marks



(OR) Any equivalent TM

$(q_0, \underline{\Delta}aabbccdd\Delta) \vdash (q_1, \Delta\underline{a}abbccdd\Delta) \vdash (q_2, \Delta A\underline{a}bbccdd\Delta)$

$\vdash (q_2, \Delta A\underline{a}\underline{b}ccdd\Delta) \vdash (q_3, \Delta AaB\underline{b}ccdd\Delta) \vdash (q_3, \Delta AaBb\underline{c}cdd\Delta)$

$\vdash (q_4, \Delta AaBbC\underline{c}dd\Delta) \vdash (q_4, \Delta AaBbCc\underline{d}d\Delta) \vdash (q_5, \Delta AaBbC\underline{c}Dd\Delta)$

$\vdash (q_5, \Delta AaBb\underline{C}cDd\Delta) \vdash (q_5, \Delta AaB\underline{b}CcDd\Delta) \vdash (q_5, \Delta A\underline{a}BbCcDd\Delta)$

$\vdash (q_5, \Delta A\underline{a}BbCcDd\Delta) \vdash (q_5, \Delta\underline{A}aBbCcDd\Delta) \vdash (q_1, \Delta A\underline{a}BbCcDd\Delta)$

$\vdash (q_2, \Delta AA\underline{B}bCcDd\Delta) \vdash (q_2, \Delta AAB\underline{b}CcDd\Delta) \vdash (q_3, \Delta AABB\underline{C}cDd\Delta)$

$\vdash (q_3, \Delta AABBC\underline{c}Dd\Delta) \vdash (q_4, \Delta AABBCC\underline{D}d\Delta) \vdash (q_4, \Delta AABBCCD\underline{d}\Delta)$

$\vdash (q_5, \Delta AABBCC\underline{D}D\Delta) \vdash (q_5, \Delta AABBC\underline{C}DD\Delta) \vdash (q_5, \Delta AABB\underline{C}CDD\Delta)$

$\vdash (q_5, \Delta AA\underline{B}BCCDD\Delta) \vdash (q_5, \Delta A\underline{A}BBCCDD\Delta) \vdash (q_5, \Delta\underline{A}ABBCCDD\Delta)$

$\vdash (q_1, \Delta A\underline{A}BBCCDD\Delta) \vdash (q_6, \Delta AAB\underline{B}CCDD\Delta) \vdash (q_6, \Delta AABB\underline{C}CDD\Delta)$

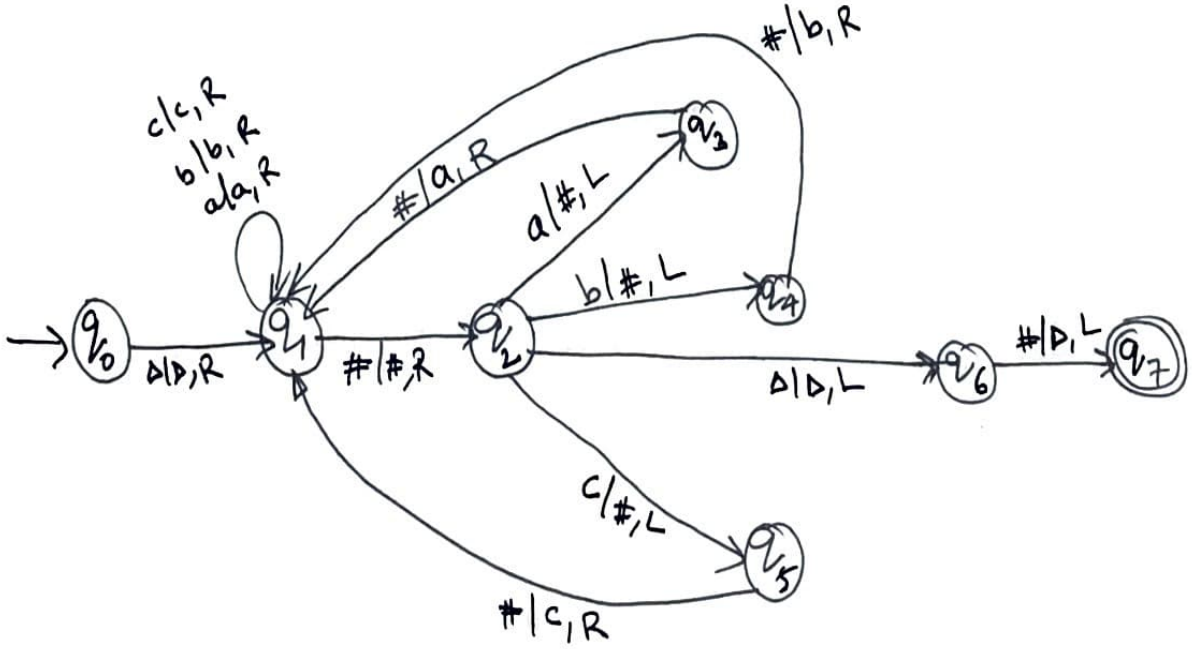
$\vdash (q_7, \Delta AABBC\underline{C}DD\Delta) \vdash (q_7, \Delta AABBCC\underline{D}D\Delta) \vdash (q_8, \Delta AABBCCD\underline{D}\Delta)$

$\vdash (q_8, \Delta AABBCCDD\underline{\Delta}) \vdash (q_9, \Delta AABBCCDD\underline{\Delta})$

Accepted.

2. TM Construction to Compute the given function $f(x,y)$ —— 6 Marks

To trace the TM for $x = bcab$ and $y = abca$ —— 4 Marks



	x						y				
Initial	Δ	b	c	a	b	#	a	b	c	a	Δ

Final	Δ	b	c	a	b	a	b	c	a	Δ	

$(q_0, \underline{\Delta} bcab\#abca\Delta) \vdash (q_1, \Delta \underline{b} cab\#abca\Delta)$

$\vdash (q_1, \Delta b\underline{c}ab\#abca\Delta) \vdash (q_1, \Delta bc\underline{a}b\#abca\Delta) \vdash (q_1, \Delta bca\underline{b}\#abca\Delta)$

$\vdash (q_1, \Delta bcab\underline{\#}abca\Delta) \vdash (q_2, \Delta bcab\#\underline{a}bca\Delta) \vdash (q_3, \Delta bcab\underline{\#}\#bca\Delta)$

$\vdash (q_1, \Delta bcaba\underline{\#}bca\Delta) \vdash (q_2, \Delta bcaba\#\underline{b}ca\Delta) \vdash (q_4, bcaba\underline{\#}\#ca\Delta)$

$\vdash (q_1, \Delta bcabab\underline{\#}ca\Delta) \vdash (q_2, \Delta bcabab\#\underline{c}a\Delta) \vdash (q_5, \Delta bcabab\underline{\#}\#a\Delta)$

$\vdash (q_1, \Delta bcababc\underline{\#}a\Delta) \vdash (q_2, \Delta bcababc\#\underline{a}\Delta) \vdash (q_3, \Delta bcababc\underline{\#}\#\Delta)$

$\vdash (q_1, \Delta bcababca\underline{\#}\Delta) \vdash (q_2, \Delta bcababca\#\underline{\Delta}) \vdash (q_6, \Delta bcababca\underline{\#}\Delta)$

$\vdash (q_7, \Delta bcababca\underline{\Delta})$

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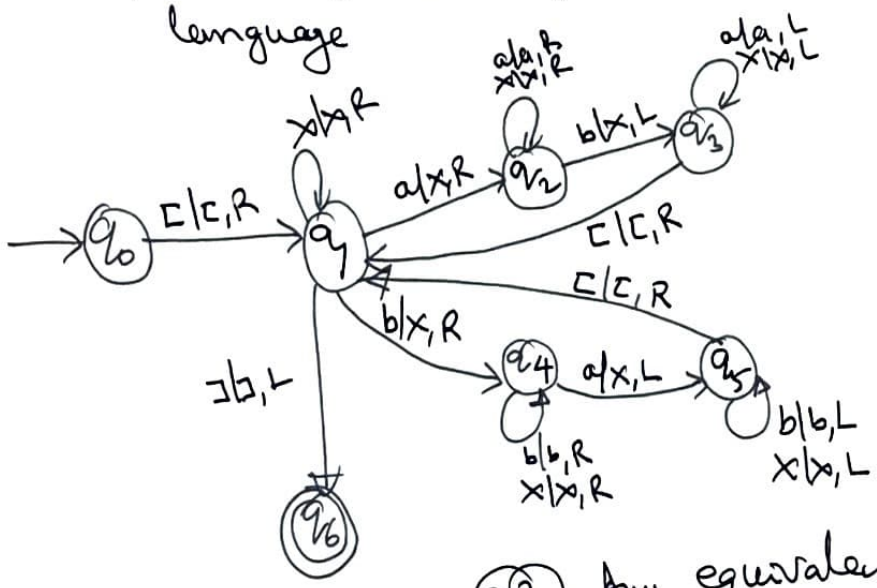
3.a) LBA definition — 1 mark

Language of LBA definition — 1 mark

LBA Construction for the given language — 4 marks



(OR) Any equivalent LBA.

3.b) Explanation of

i) Multitape TM — 1 mark

Multi track TM — 1 mark

ii) Restricted TM — 1 mark

TM with stay option — 1 mark

4.a) Definition of PCP — 2 marks

Solutions to the given PCP

15234434 — One possible solution
— Student can write any other possible solutions

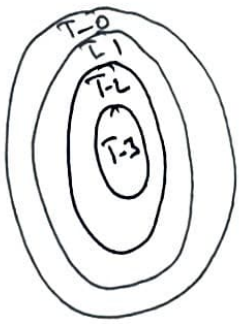
$w_1 w_5 w_2 w_3 w_4 w_4 w_3 w_4 = x_1 x_5 x_2 x_3 x_4 x_4 x_3 x_4$

10101001000100 = 10101001000100

4.b)

Type	Language	Machine	Grammar
0	REL	TM	URG
1	CSL	LBA	CSG
2	CFL	PDA	CFG
3	Reg. L	FA	Reg. Grammar

— 3 marks



— 2 marks

5.a) Definition of Unrestricted grammar — 1 mark
URG for the given language — 3 marks
Derivation for the given string aabbaabb — 2 marks

$S \to aBSAb \mid \epsilon$
$Ba \to aB$
$bA \to Ab$
$A \to a$
$B \to b$

(or) Any other equivalent grammar

$S \Rightarrow aBSAb \Rightarrow aBaBSAbAb \Rightarrow aaBBSAbAb \Rightarrow aaBBAbAb$
$\Rightarrow aaBBbAAb \Rightarrow aaBBAbAb \Rightarrow aaBBAAbb \Rightarrow aabBAAbb$
$\Rightarrow aabbAAbb \Rightarrow aabbaAbb \Rightarrow aabbaabb$

5.b) To prove $L_1 \cup L_2$ is recursively enumerable — 2 marks
To prove $L_1 \cap L_2$ is recursively enumerable — 2 marks